Egypt Real Estate AI

AI-Powered Property Valuation for the Egyptian Market

119,916 Listings | R2 = 0.7061 | 6 Governorates

Agenda

1. The Problem
2. Our Solution
3. Dataset & Model
4. Tech Stack
5. Features Overview
6. Live Demo Screenshots
7. Results & Performance
8. Competitive Analysis
9. Future Roadmap

The Problem

Difficulty in Valuing Properties

Buyers and sellers struggle to determine fair prices in Egypt.

No Standardized Valuation

Egypt lacks a reliable AI-powered valuation tool like Zestimate.

Market Opacity

Prices vary wildly across districts with no clear benchmark.

Language Barrier

Most tools are English-only, ignoring 100M+ Arabic speakers.

Our Solution

- AI-powered price estimation using CatBoost

- Trained on 119,916 real Egyptian listings

- Full Arabic (RTL) + English support

- Instant results with confidence intervals

- Comprehensive market analysis & ROI

- Dark mode + mobile-friendly design

- Free PDF reports & shareable links

- Property comparison & favorites

Key Metrics

0.7061 R2 Score	**40.52%** MAPE	**119,916** Listings
6 Governorates	**573** Districts	**881** Compounds

Dataset

119,916 Listings from 7 Data Sources

Governorate	Listings	%
Cairo	65,674	54.8%
Giza	30,374	25.3%
Matrouh	10,545	8.8%
Red Sea	6,253	5.2%
Alexandria	4,122	3.4%
Suez	2,948	2.5%

Machine Learning Model

CatBoost Regressor

Gradient Boosting on Decision Trees

Algorithm:	CatBoost Regressor
Training Set:	95,932 listings (80%)
Test Set:	23,984 listings (20%)
Features:	32 engineered features
R2 Score:	0.7061
MAPE:	40.52%

Tech Stack

Frontend	Streamlit
ML Model	CatBoost
Data Processing	Pandas, NumPy
Charts	Plotly
Maps	Folium
PDF Generation	ReportLab
Styling	Custom CSS
Deployment	Streamlit Cloud

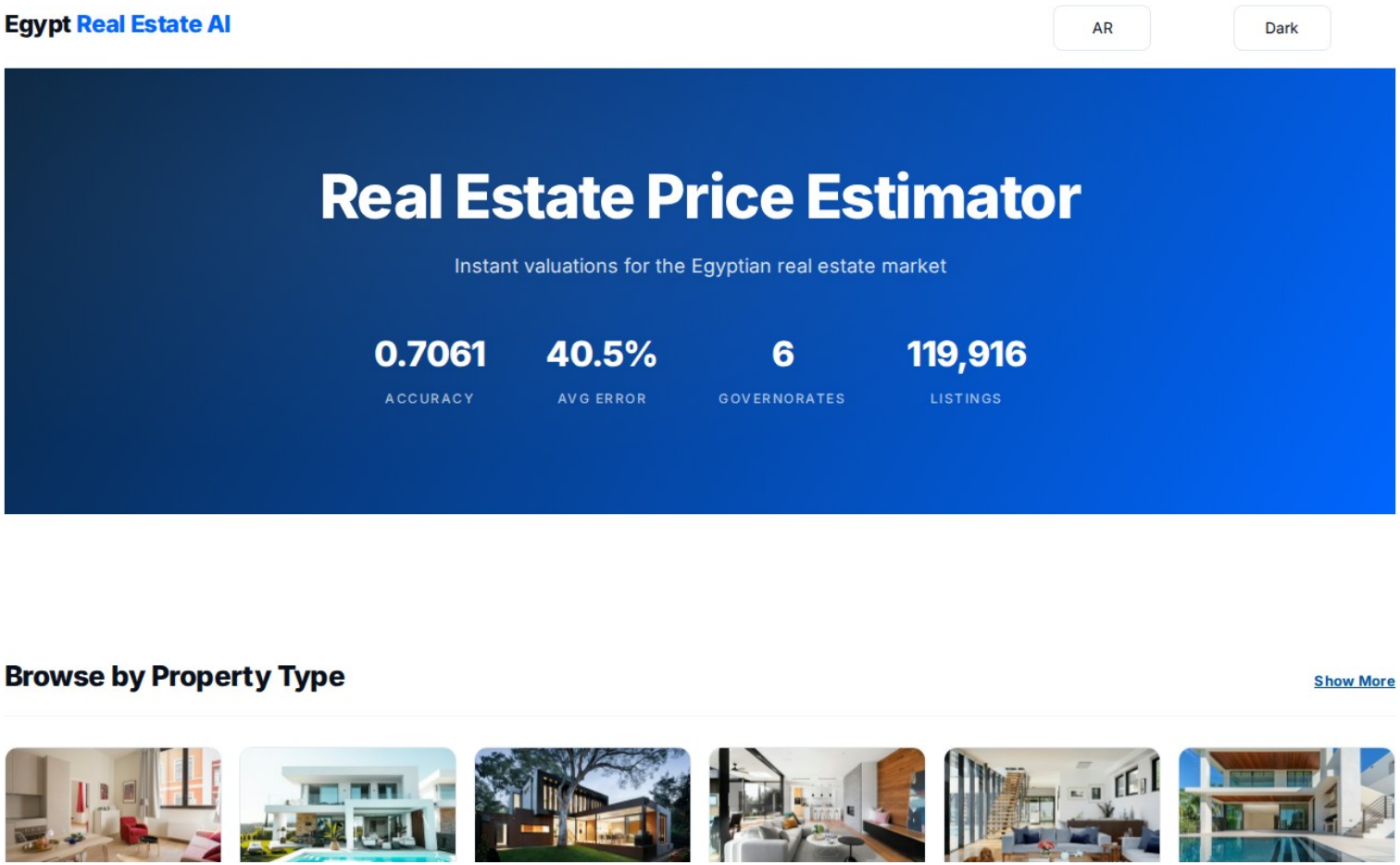
Key Features (1/2)

AI Price Estimator	5-level location + 32 features
Property Gallery	5 sample images per property type
Price History & Forecast	2020-2027 with interactive chart
Property Location	Interactive Leaflet map
Investment Analysis (ROI)	6 metrics including yield & payback
Neighborhood Insights	Schools, hospitals, transport

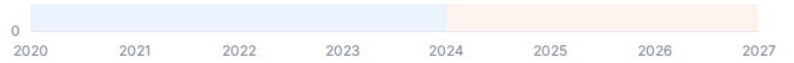
Key Features (2/2)

Market Analysis	7 interactive charts + filters
Advanced Mortgage Calculator	Pie & balance charts, amortization
Compare Properties	Side-by-side 2-4 properties
Favorites + Export/Import	Save searches & property alerts
Chat with Agent	Built-in UI for agent contact
Dark Mode + Arabic RTL	Full bilingual support

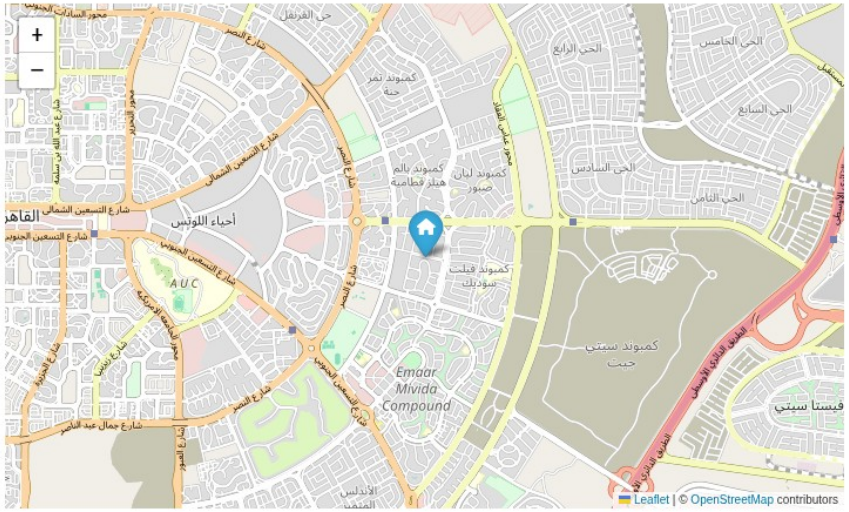
Live Demo - Homepage



Live Demo - Estimation Result



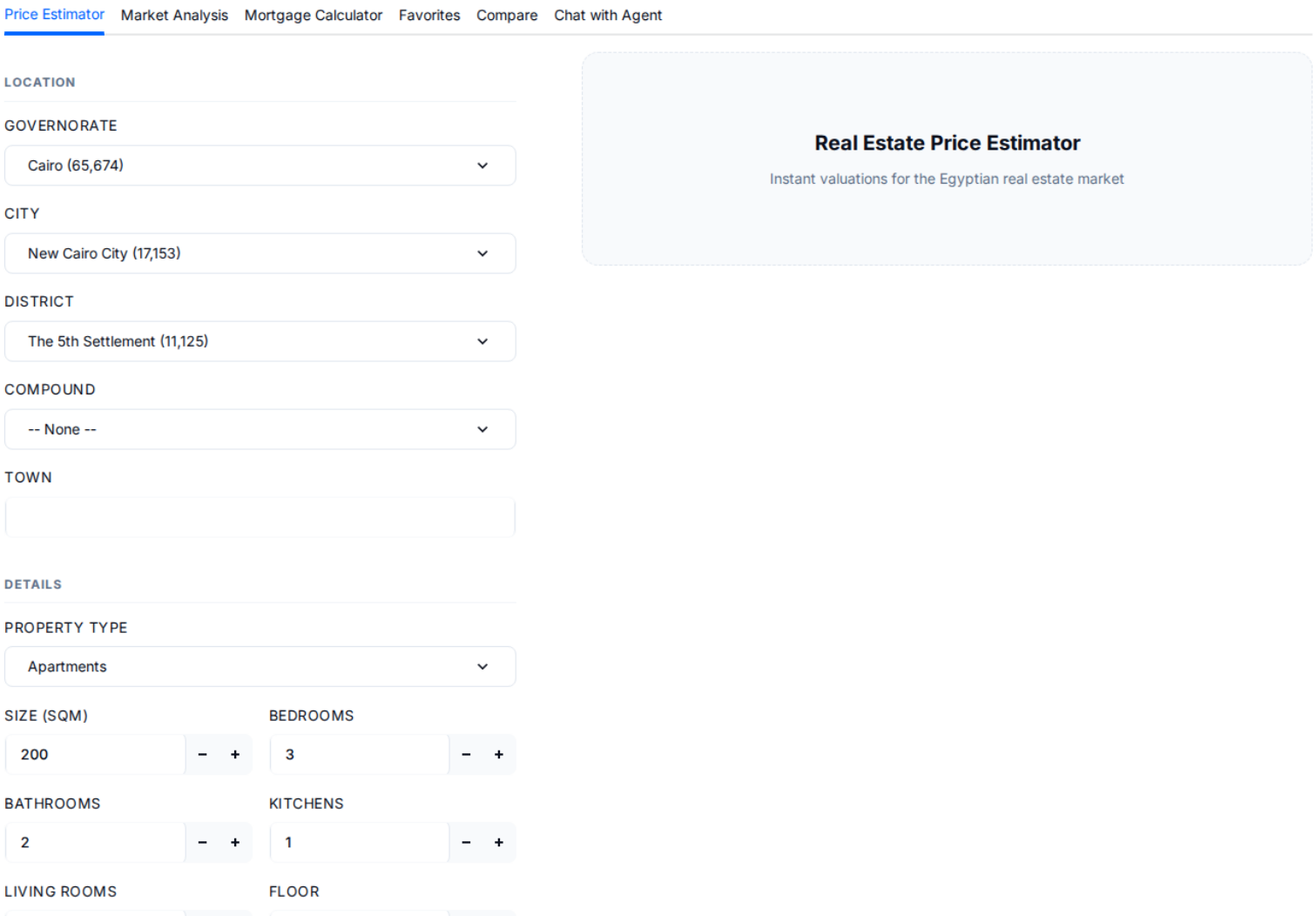
Property Location



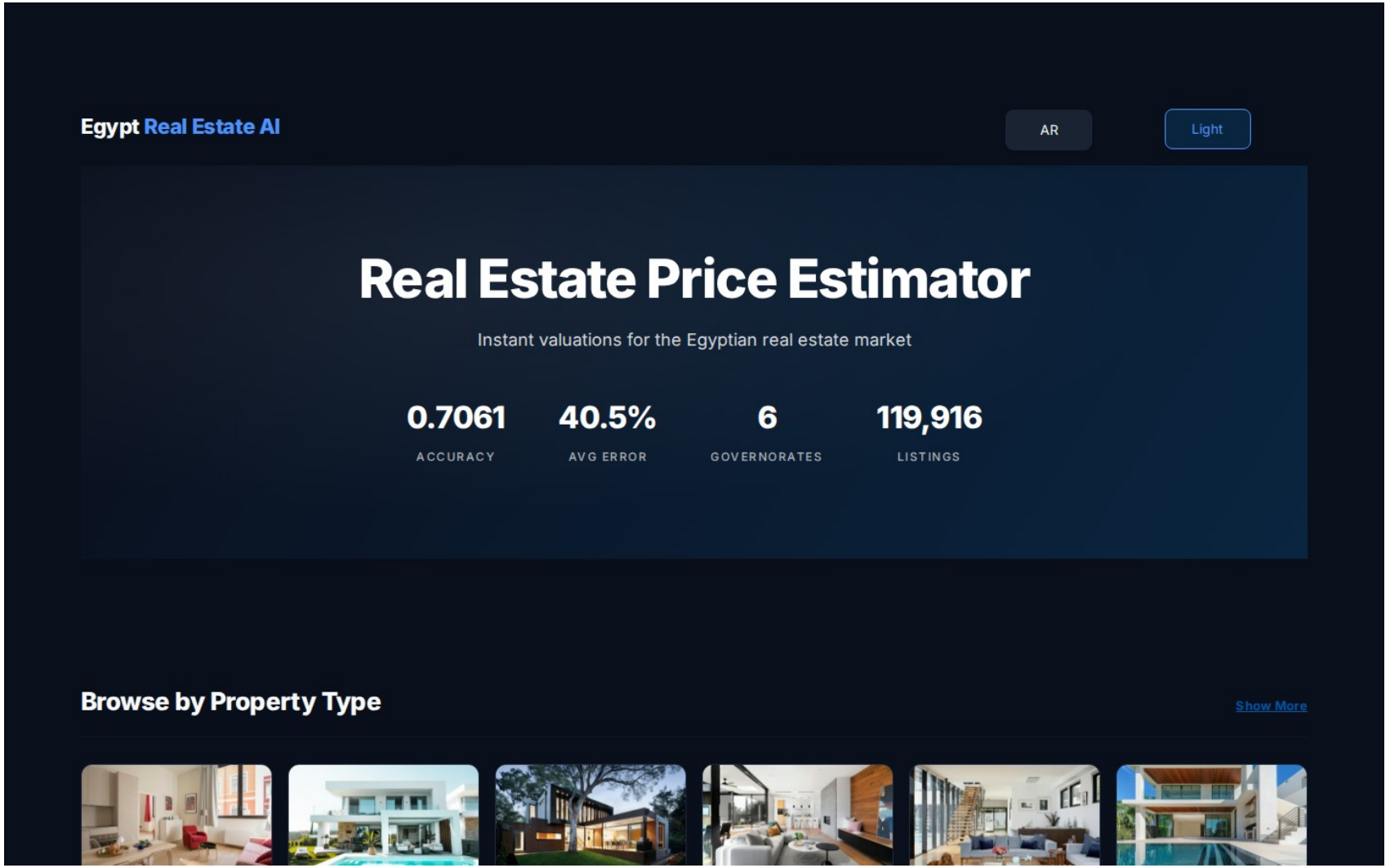
Investment Analysis

MONTHLY RENT	ANNUAL YIELD	PAYBACK PERIOD
41,394 EGP	5.50 %	18.2 years

Live Demo - Property Types



Live Demo - Dark Mode



Live Demo - Investment Analysis

1 − +

3 − +

FEATURES

FINISHING

Core & Shell

FURNISHED

Unfurnished

VIEW

No Specific View

DEVELOPER

SELLER TYPE

Broker

COMPLETION STATUS

Ready

ADDITIONAL FEATURES

Select available features

Garden	Pool
Garage	Security 24/7
Gym	Elevator
Balcony	View
Mall	Mosque
Sports Club	Sea Front

RANGE

5.4M - 12.7M

CONFIDENCE

71%

DATA QUALITY

• High

5221 in this district

VS. MARKET (DISTRICT)

11%

below market

TARGET PRICE (EGP)

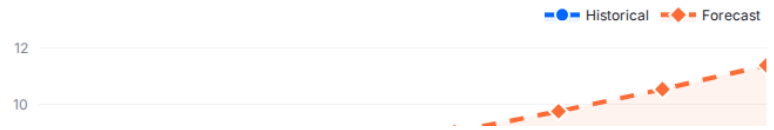
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Set Alert

Save | PDF Report | Share

Price History & Forecast

1 YEAR AGO	CURRENT PRICE	PRICE IN 1 YEAR	PRICE IN 3 YEARS
8.36M	9.03M	9.75M	11.38M
M EGP	M EGP	M EGP	M EGP
8.0% since last year	Current	+8.0% forecast	+26.0% forecast



Results & Performance

R2 Score	**0.7061**	70.6% variance explained
MAPE	**40.52%**	Average error margin
Sample Size	**119,916**	Real listings from Egypt
Coverage	**6**	Governorates with data
Inference Time	**< 2s**	Instant predictions
Accuracy	**High**	On Cairo, Giza, Alexandria

Competitive Analysis

How we compare to industry leaders:

Feature	Zillow	Nawy	Aqarmap	Ours
AI Estimate	Yes	No	No	Yes
Price History	Yes	Yes	No	Yes
ROI Analysis	Yes	No	No	Yes
Arabic Support	No	Yes	Yes	Yes
Dark Mode	Yes	Yes	Yes	Yes
PDF Reports	Yes	No	No	Yes

System Architecture

User Interface

Streamlit + Custom CSS (Nawy Theme)

Business Logic

Python: Prediction, Charts, Maps, PDF

ML Model

CatBoost Regressor (12 MB)

Data Layer

119,916 listings + Geography hierarchy

Future Roadmap

Phase 1 (Done)

- AI Price Estimator
- Property Gallery
- Price History
- ROI Analysis

Phase 2 (Done)

- Market Analysis
- Mortgage Calculator
- Compare
- Chat UI

Phase 3 (Next)

- Real Property Photos
- Property Details Page
- 3D Tours
- Mobile App

Key Takeaways

- First AI-powered bilingual property valuation for Egypt
- Trained on 119,916 real listings (largest dataset in market)
- R2 = 0.7061 - comparable to international standards
- Full Arabic support - targeting 100M+ users
- Competes with Zillow, Nawy, Aqarmap on core features
- Ready for commercial deployment

Thank You

Egypt Real Estate AI

Live Demo: smart-house-price-predictor-77.streamlit.app

GitHub: github.com/qamarsobhy7-source/Smart-House-Price-Predictor

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